

Huiwen Tan

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Education

The Chinese University of Hong Kong, Hong Kong, China

Aug. 2025 – Jun. 2027

- Master of Philosophy in Chemistry
- **Coursework:** Supramolecular Chemistry; Organometallic Chemistry & Catalysis; From Computational Physics to AI

The University of Queensland, Australia

Feb. 2020 – Dec. 2024

- Bachelor of Advanced Science (Honours) in Chemistry
- **Honours Thesis:** Modelling spin-crossover lattices with machine-learned force fields
- **Awards and Honours:** Honours Class I; Dean's Commendation for Academic Excellence (2020, 2024); Summer Scholarship for Summer Research Program (2022/23)

Publications

Sun, J.†; **Tan, H.†**; Zhang, Y.; Zhang, Y.-Q.; Huang, Z.; Zhou, W.-Q.; Zou, T.; Li, M.; Zhang, X.; Zhang, Q.-W., Enzyme-inspired multipoint anchoring enables an asymmetric thio-Povarov reaction. Submitted to *Science*.

Vummaleti, S. V. C.†; **Tan, H.†**; Liu, J.; Zhang, X., CO₂ hydrogenation to methanol catalyzed by Mn-SNS complex: role of *fac/mer* isomerism on catalytic reactivities. *ACS Catal.*, **2026**. DOI: 10.1021/acscatal.6c03391.

Huang, Z.; **Tan, H.**; Cui, R.; Hu, Y.; Zhang, S.; Jia, J.; Zhang, X.; Zhang, Q.-W., Regiodivergent hydrophosphination of bicyclo[1.1.0]-butanes under catalyst control. *Nat. Commun.*, **2025**, 16, 6216. DOI: 10.1038/s41467-025-61415-8.

Zhang, X.; **Tan, H.**; Liu, J.; Li, Z.; Wang, L.; Chen, B. W. J., CHEMSMART: chemistry simulation and modeling automation toolkit for high-efficiency computational chemistry workflows. *arXiv*, **2025**, arXiv:2508.20042.

Manuscripts

Tan, H.; Zhang, X., Automated construction of publication-ready Gibbs free energy profiles from quantum chemistry output files.

Tan, H.; Zhang, X., CHEMSMART database: a lightweight, provenance-aware relational database for molecular quantum chemistry.

Tan, H.; Natt, N.; Powell, B. J., Spin crossover dynamics from first principles: effects of ligand substitution and solvent environment.

Skills

Computational Chemistry: VASP, Gaussian, ADF, xTB, ORCA, Quantum ESPRESSO, CREST

Scientific Computing & ML: Python (NumPy, SciPy, Pandas, RDKit, TensorFlow, scikit-learn, Matplotlib, ASE, Phonopy, pymatgen), C/C++ , MATLAB, R, Mathematica

Software & Data Engineering: SQL (SQLite), Git/GitHub, TypeScript, HTML

Research Experiences

Computational Chemistry Workflow and Software Development Jan. 2025 – Present

- Developed automated computational workflows in CHEMSMART for quantum chemistry calculations, thermochemistry analysis, conformational sampling, and database construction
- Developed a provenance-aware relational database for assembling and querying molecular quantum-chemistry calculations and associated metadata
- Developed a quasi-RRHO-based thermochemistry workflow for automated thermal corrections and construction of Gibbs free energy profiles from Gaussian, ORCA, and xTB outputs
- Implemented xTB and CREST workflows for automated geometry optimization, frequency calculations, and conformational exploration

On-the-Fly Machine Learning Force Fields for Solid-State Spin Crossover Materials Jan. 2024 – Dec. 2024

- Developed on-the-fly machine learning force fields using Bayesian linear regression during *ab initio* molecular dynamics simulations
- Achieved DFT-level accuracy in predicting spin crossover (SCO)-related properties, including equilibrium structures across spin states, enthalpy differences, and phonon density of states for $[\text{Fe}(\text{ptz})_6](\text{BF}_4)_2$
- Proposed a machine learning-driven molecular dynamics scheme that captures thermal effects in the SCO phenomenon via quasi-harmonic approximation and reproduces the experimental SCO behavior through mixing between spin states

Density Functional Theory (DFT) Study of the Light-Induced Excited Spin-State Trapping Effect in Spin Crossover Materials Nov. 2022 – Jun. 2023

- Investigated SCO dynamics for phosphorene complexes $[\text{Fe}(\text{dppen})_2\text{X}_2]$ ($\text{X} = \text{Br}$ or Cl) *in vacuo* and in COSMO solvent environments from DFT calculations
- Devised a parameter-free computational scheme to model the all-atom free energy surface and kinetic barriers associated with spin-state interconversions
- Reproduced the experimentally observed trends in the influence of relaxation kinetics of the trapped excited state

Stereoselective Reduction of 5-C-bromo-D-glucuronyl β -fluoride with the Mediation of Tributyltin Hydride: A DFT Investigation Jul. 2022 – Nov. 2022

- Identified competing transition-state conformers and determined stereoselectivity through conformational sampling and DFT barrier calculations, revealing a transition-state gauche effect as the origin of the reaction mechanism
- Benchmarked product isomer ratios L-ido:D-gluco using 7 exchange-correlation functionals, yielding computational results between 1:0.03 and 1:0.06 close to the experimental value of 1:0

Work Experiences

Teaching Assistant, The Chinese University of Hong Kong – Hong Kong, China Sep. 2025 – Present

Casual Academics, The University of Queensland – Brisbane, Australia Feb. 2024 – Nov. 2024

Administrative Intern, Beijing Genomics Institute – Shenzhen, China Oct. 2021 – Dec. 2021

Testing Technician, IMPAQ Testing Technology Co., LTD – Shenzhen, China Aug. 2021 – Oct. 2021